



Database security unit 5 question bank

Database Security And Privacy (SRM Institute of Science and Technology)



Scan to open on Studocu

SRMINSTITUTE OF SCIENCE AND TECHNOLOGY

Ramapuram Campus, BharathiSalai, Ramapuram, Chennai-600089

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

QUESTION BANK

SUBJECT: 18CSE455E/ DATABASE SECURITY AND PRIVACY

SEM/YEAR: VII/IV

Course Outcomes

CO1: Acquire the knowledge of information system and information security

CO2: Able to manage the security of information system as well as database

CO3: Able to design and develop the security model in database

CO4: Able to implement VPD in various

CO5: Able to audit the database activities, users, security

CO6: Apply the security mechanism in PPDM using various algorithms

UNIT-V			
Privacy Preserving Data Mining Techniques: Introduction, Privacy Preserving Data Mining Algorithms, General Survey, Randomization Methods, Group Based Anonymization, Distributed Privacy Preserving Data Mining, Curse of Dimensionality, Application of Privacy Preserving Data Mining, Case study: on PPDM			
PART-A (Multiple Choice Questions)			
Q. No	Questions	Course Outcome	Competence BT Level
1	PPDM means (ans a) a. Privacy preserving data mining b. Privacy producing data mining c. Privacy preserving data modeling d. Privacy producing data modeling	CO-6	BT1
2	In which method noise is added to data (ans a) a. The randomization method b. The k-anonymity model and l-diversity c. Distributed privacy preservation d. Downgrading Application Effectiveness	CO-6	BT1
3	In which model we reduce the granularity of data representation with the use of techniques such as generalization and suppression (ans b) [PG.NO-20][L2] a. The randomization method b. The k-anonymity model and l-diversity c. Distributed privacy preservation d. Downgrading Application Effectiveness	CO-6	BT1
4	Which model is used to maintain privacy for each individual entity, while deriving aggregate results over the entire data? (ans c) a. The randomization method	CO-6	BT1

	b. The k-anonymity model and l-diversity c. Distributed privacy preservation d. Downgrading Application Effectiveness		
5	Association rule mining, classification or query processing privacy violations can be prevented by (ans d) a. The randomization method b. The k-anonymity model and l-diversity c. Distributed privacy preservation d. Downgrading Application Effectiveness	CO-6	BT1
6	In which case the attacker knows some linearly independent collection of records, and their corresponding perturbed version (ans a) a. Known Input-Output Attack b. Known Input Attack c. Known Output Attack d. Known Sample Attack	CO-6	BT1
7	In which case the attacker has a collection of independent data samples from the same distribution from which the original data was drawn.(ans d) a. Known Input-Output Attack b. Known Input Attack c. Known Output Attack d. Known Sample Attack	CO-6	BT2
8	The values across different records are swapped in order to perform the privacy-preservation in (ans b) a. Data mapping b. Data swapping c. Data swapping and mapping d. Data ordering	CO-6	BT1
9	k-anonymity techniques uses (ans c) a. generalization b. suppression c. generalization and suppression d. randomization	CO-6	BT1
10	In which attack, all the values for a sensitive attribute within a group of k records are the same (ans c) a. Background Knowledge Attack b. Known Input-Output Attack c. Homogeneity Attack d. randomization	CO-6	BT2
11	The attack where the adversary can use an association between one or more quasi-identifier attributes with the sensitive attribute in order to narrow down possible values of the sensitive field further is called as a. Background Knowledge Attack b. Known Input-Output Attack c. Homogeneity Attack d. randomization	CO-6	BT1
12	The t-closeness Model is the enhancement of(ans a) a. l-diversity model b. k-anonymity c. both a and b d. None of the above	CO-6	BT2

13	The individual records are spread out across multiple entities, each of which have the same set of attributes are in (ans b) a. vertically partitioned b. horizontally partitioned c. Diagonally partitioned d. Randomly partitioned	CO-6	BT2
14	The individual entities may have different attributes (or views) of the same set of records are in (ans a) a.vertically partitioned b. horizontally partitioned - c.Diagonally partitioned - d.Randomly partitioned	CO-6	BT1
15	In association rule hiding if the entry for a given transaction is modified to a different value then it is called as a. Blocking b. Aborting c. Distortion d. Hiding	CO-6	BT2
16	Background knowledge referred to (ans a) a. Additional acquaintance used by a learning algorithm to facilitate the learning process b. A neural network that makes use of a hidden layer c. It is a form of automatic learning. d. A neural network that not makes use of a hidden layer	CO-6	BT2
17	Classification is (ans a) a. A subdivision of a set of examples into a number of classes b. A subdivision of a set of examples into a number of functions c. A measure of the accuracy, of the classification of a concept that is given by a certain theory d. The task of assigning a classification to a set of examples	CO-6	BT1
18	Query tools are (ans c) a. A reference to the speed of an algorithm, which is quadratically dependent on the size of the data b. Attributes of a database table that can take only numerical values. c. Tools designed to query a database. d. Attributes of a database table that can not take only numerical values	CO-6	BT1
19	Inductive learning is (ans c) web a. Machine-learning involving different techniques b. The learning algorithmic analyzes the examples on a system atic basis and makes incremental adjustments to the theory that is learned c. Learning by generalizing from examples d. Learning by generalizing from designs.	CO-6	BT1
20	Which one is not suitable fork-anonymity techniques uses in (ans d)web a. generalization b. suppression	CO-6	BT1

	c. generalization and suppression d. randomization		
21	Cryptographic algorithms are based on mathematical algorithms where these algorithms use _____ for a secure transformation of data. (ans a) web a) secret key b) external programs c) add-ons d) secondary key	CO-6	BT1
22	Which one of the following is a cryptographic protocol used to secure HTTP connection? (ans b) web a) stream control transmission protocol (SCTP) b) transport layer security (TSL) c) explicit congestion notification (ECN) d) resource reservation protocol.(RRP)	CO-6	BT1
23	The individual records are spread out across multiple entities, each of which have the same set of attributes are in (ans b) [PG.NO-28][L1] a. vertically partitioned b. horizontally partitioned c. Diagonally partitioned d. Randomly partitioned	CO-6	BT1
24	Key to represent relationships between tables is called (ans b) a. Primary key b. Foreign key c. Secondary key d. Super key	CO-6	BT1
25	Euclidean distance measure is (ans C) a. A stage of the KDD process in which new data is added to the existing selection. b. The process of finding a solution for a problem simply by enumerating all possible solutions according to some pre-defined order and then testing them c. The distance between two points as calculated using the Pythagoras theorem d. The process of finding a solution for a problem simply by enumerating all possible solutions	CO-6	BT1
PART B (4 Marks)			
1	How the data mining techniques are used to preserve privacy in database.	CO-6	BT1
2	What are Privacy Quantification	CO-6	BT1
3	What are privacy preserving algorithms available in data mining? Explain any one.	CO-6	BT2
4	What is Randomization? Explain in brief	CO-6	BT1
5	What is group based Anonimization?	CO-6	BT4

6	What is distributed privacy preserving?	CO-6	BT2
7	What are the applications of privacy preserving data mining?	CO-6	BT6
8	Give the general survey about randomization method.	CO-6	BT3
9	Why the data mining techniques preferred for preserving privacy?	CO-6	BT1
10	Define Semi-honest Adversaries.	CO-6	BT1
11	State Multiplicative Perturbation	CO-6	BT2
12	State additive Perturbation	CO-6	BT1
13	List out the Malicious Adversaries.	CO-6	BT4
14	Analyse the l-diversity Method.	CO-6	BT1
PART C (12 Marks)			
1	List the privacy preserving algorithms available in data mining? Explain all in detail.	CO-6	BT1
2	Explain in detail about Randomization method.	CO-6	BT3
3	Compare and contrast the Distributed Algorithms over horizontally and vertically	CO-6	BT3
4	Explain the Distributed Algorithms for k-Anonymity.	CO-6	BT4
5	Examine the Adversarial Attacks on Randomization.	CO-6	BT5

Note:

1. **BT Level** – Blooms Taxonomy Level

2. **CO – Course Outcomes**

BT1 –Remember BT2 – Understand BT3 – Apply BT4 – Analyze BT5 – Evaluate BT6 – Create